Peiran Xu

peiran@ucla.edu ⋄ peiranxu.github.io

Education

University of California, Los Angeles Los Angeles, CA, USA
Ph.D., Interests: LLM Post-Training, AI Agents and AI for Science Sep 2023 - present

Shanghai Jiao Tong University Shanghai, China
Bachelor of Engineering Sep 2019 - June 2023

Selected Publications

- ***Peiran Xu****, *Zhuohao Li**, *Xiaoying Xing*, *Guannan Zhang*, *Debiao Li*, *Kunyu Shi*: Hybrid Reward Normalization for Process-supervised Non-verifiable Agentic Tasks. Under review. [arXiv]
- ***Peiran Xu****, *Zeyu Wang**, *Jieru Mei*, *Alan Yuille*, *Liangqiong Qu*, *Cihang Xie*, *Yuyin Zhou*: FedConv: Enhancing Convolutional Neural Networks for Handling Data Heterogeneity in Federated Learning. **(TMLR'24)** [arXiv]
- *Jianshu She**, *Zhuohao Li**, *Zhemin Huang*, *Qi Li*, ***Peiran Xu***, *Haonan Li*, *Qirong Ho*: Hawkeye: Efficient Reasoning with Model Collaboration. **(COLM'25)** [arXiv]
- *Yifan Gao*, ***Peiran Xu***, *Yimeng He*, *Haoran Li*, *Ziyang Long*, *Debiao Li*: Shared Gaussian Geometry for Zero-Shot MRI Spatial Resolution Harmonization. **(MICCAI'26)**

Industry Experience

Alibaba Group Sunnyvale, CA, USA
Research Intern, advisor: Kunyu Shi Jun 2025 - Dec 2025

- Developed a principle-based Process Reward Model (PRM) that evaluates intermediate reasoning steps against general principles and sample-specific rubrics, enabling interpretable and principle-aligned supervision.
- Designed and implemented ReNorm, a unified reward normalization framework integrates and balances process and outcome rewards in RL training, effectively stabilizing training and preventing collapse in RL optimization.

Snowflake Inc Menlo Park, CA, USA
Research Intern, manager: Zhewei Yao Feb 2026 - present

- Developed the web-search part of hybrid deep-research benchmark designed to evaluate agents' ability to retrieve, synthesize, and reason over both structured database records and open-web information.

Research Experience

Cedars Sinai Los Angeles, CA, USA
Research Assistant Jul 2024 - present

- Developed weakly supervised method to generate T1/T2 mappings, leveraging MR physics equations for self-supervised pretraining and fine-tuning with minimal data. Validated clinical utility on a large-scale internal dataset. Paper under review.
- Developing a multimodal evaluation framework for AI-agent radiology report generation, including rubric-based judge reward model training and a novel radiologist-annotated benchmark for clinical correctness and report quality assessment.

Department of Computer Science, John Hopkins University Baltimore, MD, USA
Research Intern, advisor: Prof. Alan Yuille, Prof. Cihang Xie and Prof. Yuyin Zhou Jun 2022 - Sep 2022

- Research in Designing Robust Convolution Neural Networks for Federated Learning (FL)
- Conducted comprehensive investigation of Vision Transformer components and successfully integrated key elements into CNNs, significantly enhancing model performance in tackling heterogeneous data for FL in vision tasks
- Proposed a robust CNN architecture that achieved state-of-art results across various FL benchmarks, advancing the understanding of deep learning network architecture and setting a stronger baseline for future FL research.

Shanghai Jiao Tong University Shanghai, China
Research Intern Oct 2021 - May 2023

- **Efficient Endoscopic Video Super-Resolution:** Overcame limited computing resources on edge devices by optimizing lightweight network, and using ONNX and TensorRT for deployment to enable real-time inference
- **Automatic Detection of Cardiac Cycle:** Developed optical-flow-based video understanding neural network to distinguish cardiac phases in coronary angiography. Selected as national student project for excellence

Services & Skills

Reviewers & AE: TMLR'25, AIIM, Usenix Security